

Exercise Setup Instructions

This document will explain how to setup your computer for the exercises. We recommend using Conda (Miniconda is sufficient) to create the course environment, as explained in Section 1. Alternatively, you can use Python's built-in `venv`, as explained in Section 2. In Sections 3 and 4 we give short introductions to PyCharm and Visual Studio Code, two editors you can use for the exercises. We also provide a video tutorial in TUWEL where we go through the steps to set up your system for the course.

Supported operating systems for the libraries we are using:

- Windows 10+ (64 bit)
- Ubuntu 22.04+
- macOS with a compatible Open3D wheel for your version and processor architecture

The recommended Python version for both installation methods is **3.12.x**. Package availability also depends on processor architecture; see the [Open3D installation requirements](#). Exercise 3 uses the separate PyTorch/notebook environment described in its assignment.

1 Conda

Conda is a package management system and environment management system. It enables installing required libraries in a separate virtual environment. This way, it won't interfere with any installed existing environments. You can either install the full fletched Anaconda or its smaller variant Miniconda. For our use-case Miniconda is sufficient. You can use either the GUI or command line version for Python 3. Installation instructions for Windows, macOS and Linux can be found [here](#).

After installing conda you can use the environment file to setup the environment using the terminal by calling (replace the `/path/to/`)

```
conda env create -f /path/to/mv_environment.yml

# Check if the environment was installed successfully by calling
conda env list

# Activate the environment by calling
conda activate machinevision

# In this environment you should be able to execute the main file of the
# first exercise
python /path/to/exercise1/main.py
```

2 Alternative: venv and pip

Conda remains the recommended setup. If you prefer `venv`, first install **Python 3.12** from python.org or your operating system's package manager. Unlike Conda, `venv` does not install Python itself. Use `requirements.txt` from the same course setup folder as `mv_environment.yml`.

On Linux or macOS, open a terminal in your working directory:

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r /path/to/requirements.txt
```

On Ubuntu, install the matching `python3.12-venv` system package if Python reports that virtual-environment support is missing.

On Windows, use Command Prompt (replace the requirements path):

```
py -3.12 -m venv .venv
.venv\Scripts\activate.bat
python -m pip install --upgrade pip
python -m pip install -r C:\path\to\requirements.txt
```

Use either the Conda instructions or these `venv` instructions for an environment; you do not need to install both sets of requirements.

After either installation method, check the imports:

```
python -c "import numpy, scipy, matplotlib, sklearn, cv2, PIL,
           pyquaternion, open3d; print('Imports OK')"
```

Then run the first exercise to check that image windows open correctly. Open3D visualization also requires working graphics drivers. For a `venv` in PyCharm, select an existing local interpreter at `.venv/bin/python` (Linux/macOS) or `.venv/Scripts/python.exe` (Windows).

3 PyCharm

An integrated development environment (IDE) provides many advantages, such as debugging, code completion, syntax highlighting, refactoring, etc. We suggest to use the cross-platform IDE PyCharm by JetBrains and provide you with a short introduction here. Of course, you can use any other IDE or text editor of your choice as well.

PyCharm is available in two variants, the free Community edition and a Professional edition. The Professional version offers more features and you are eligible to a free student license with your TU Wien email [here](#). We recommend the Professional version since it offers additional features such as a scientific view and profiling. However, the Community edition is also sufficient for this course. Download the program [here](#) and follow the provided instructions to install it.

Start PyCharm and open the folder of the first exercise. To setup the correct Python interpreter, go to

Settings > Python > Interpreter

Press **Add Interpreter** and choose **Add local interpreter**. Check the checkbox **Select existing** and in the drop-down menu select **Conda**. In the **Environment** drop-down menu, you should find the created **machinevision** environment. If it is not there, you can add the location manually. You can find the location of an environment by calling `conda env list` in your terminal. After these steps, you should be able to execute the `main.py` file by right clicking it in PyCharm and pressing **Run**. The first time you run the program, a window might popup where you choose the **machinevision** environment in the **Interpreter** field. You can find the official PyCharm tutorial for conda environments [here](#).

If you did the alternative setup using `venv`, the steps are very similar and you can find detailed instructions [here](#).

4 Visual Studio Code

Visual Studio Code (VS Code) by Microsoft is a free, cross-platform alternative to PyCharm. With its Python extension, it provides code completion, syntax highlighting, debugging, and an integrated terminal. Both editors are suitable for the exercises, so you can choose the one you prefer. However, we recommend PyCharm for this exercise as it is a bit easier to setup, while VS Code is more flexible and supports other programming tasks.

Download VS Code [here](#) and follow the installation instructions for your operating system. Start VS Code and open the folder of the first exercise using **File > Open Folder**. Open the Extensions view from the sidebar, search for **Python**, and install the extension published by Microsoft. Create the course environment using the Conda or `venv` instructions above before selecting it in the editor.

To select the correct Python interpreter, open the Command Palette using **Ctrl+Shift+P** on Windows/Linux or **Cmd+Shift+P** on macOS, then search for:

Python: Select Interpreter

Choose the `machinevision` Conda environment, or your `.venv` environment if you used the alternative setup. If it is not listed, use **Enter interpreter path** to select its Python executable manually. For Conda, you can find the environment directory by running `conda env list` in your terminal; the executable is `python.exe` inside that directory on Windows, or `bin/python` on Linux/macOS. For `venv`, select `.venv/Scripts/python.exe` on Windows or `.venv/bin/python` on Linux/macOS.

After selecting the interpreter, open `main.py` and choose **Run Python File in Terminal** using the play button in the top-right corner of the editor. The program runs with the selected Python environment, and its output appears in the integrated terminal. For debugging, set a breakpoint by clicking next to a line number and choose **Python Debugger: Debug Python File** from the Command Palette. You can then step through your program and inspect variable values. Further instructions are available in the [official Python tutorial for VS Code](#).

For the Jupyter notebook in exercise 3, also install Microsoft's **Jupyter** extension. Open the notebook and use the kernel selector in the top-right corner to choose the separate exercise-3 environment described in that assignment.